

A. Blurry Image



For this one, I used a standard embossment kernel

$$\begin{bmatrix} -2 & -1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

Which I changed the middle part into 0.12

$$\begin{bmatrix} -2 & -1 & 0 \\ -1 & 0.12 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

which makes the sum of the kernel that's **originally 1 into 0.12**

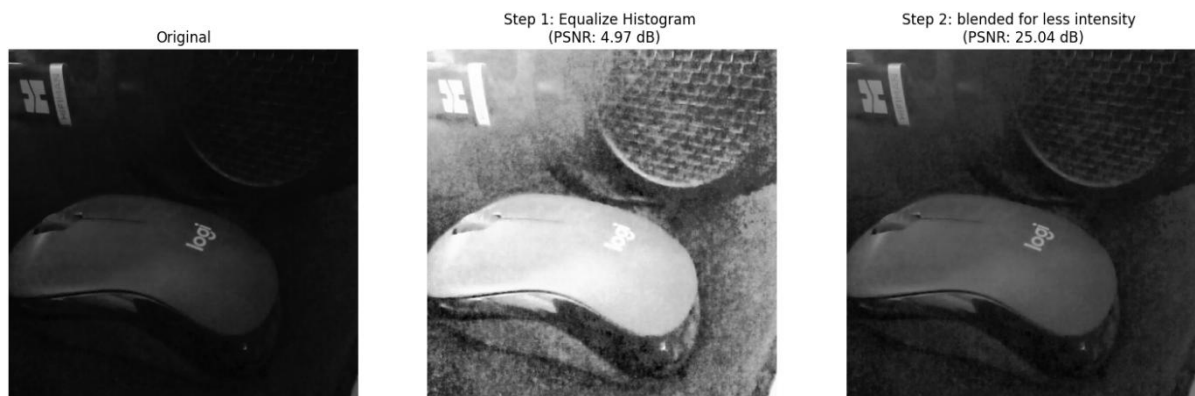
$(-2 + -1 + 0 + -1 + 1 + 1 + 0 + 1 + 2) = 1$ **this is before**

$(-2 + -1 + 0 + -1 + 0.12 + 1 + 0 + 1 + 2) = 0.12$ **this is after**

making the image darker than what the embossment kernel is supposed to look like. After that, I applied histogram equalization from opencv, to enhance the contrast of the image by changing its distribution of its pixel intensity, to make the final result like that.

As we can see, the PSNR jumped from 9.32 dB into 10.05, which might not be that significant, but the final image looks brighter than just the embossment filter I tweaked, and a lot sharper than the original image, making the blurry text look somewhat more legible.

B. Dark Image

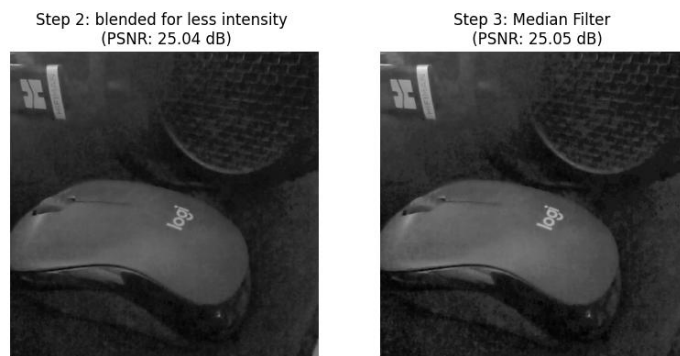


For this one, I use histogram equalization from opencv to enhance the contrast of the image by changing its distribution of its pixel intensity, then because it looks too much, I used blending

```
alpha = 0.9  
beta = 1.0 - alpha  
blended = cv2.addWeighted(img, alpha, filtered, beta, 0)
```

which reduces the histogram equalization's intensity by 90 percent, making it less harsh.

Now the final image does look salt noisy, which I tried to fix by using median filtering, but unfortunately, it only made the image more noisy, as we can see here:



And I also tried applying gaussian blur too, but for this one, it doesn't make the noise worse, only that it has little to no difference

Step 2: blended for less intensity
(PSNR: 25.04 dB)



Step 3: Gaussian Blur
(PSNR: 25.19 dB)

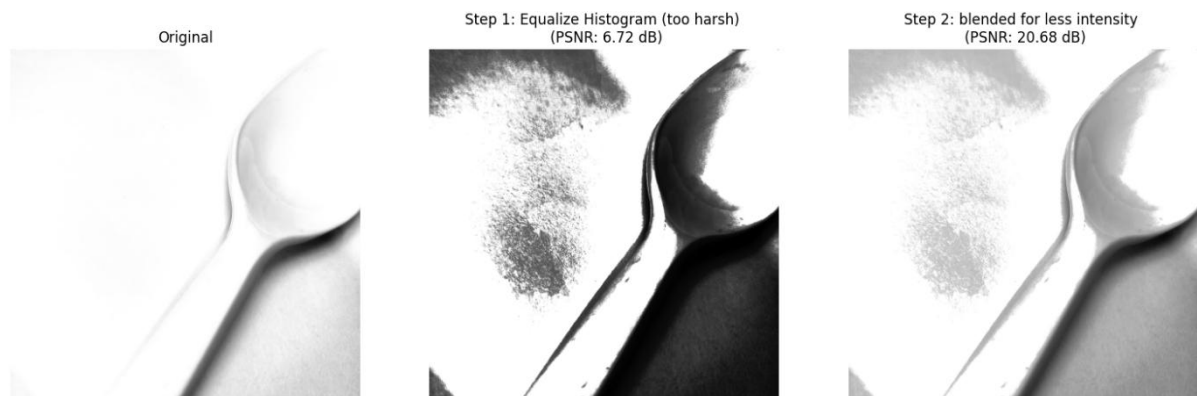


so I decided to just remove the median filtering and gaussian blur and keep the final blended one as it is.

The PSNR jumped from 4.97 dB to 25.04 dB after the blending, which proves that reducing its intensity is the right way to go. It keeps the histogram equalization's improvement, without it looking too harsh and different from the original image.

The final image looks a lot less dark, making the edges of the mouse and the headphone much more visible, instead of the original one that looks as if the edges melts into the dark background.

C. Bright Image



For this one, I use histogram equalization from opencv to enhance the contrast of the image by changing its distribution of its pixel intensity, then like last time, because it looks too harsh, I used blending

```
alpha = 0.8  
beta = 1.0 - alpha  
blended = cv2.addWeighted(img, alpha, filtered, beta, 0)
```

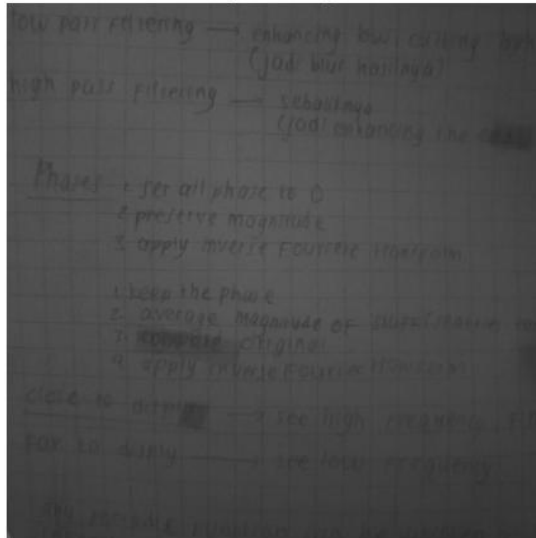
which reduces the intensity by 80 percent in this image (on the previous image it's 90). Because I think 80 percent works better for this image.

The PSNR jumped from 6.72 dB to 20.68 dB, proving that the 80 percent blending is indeed useful.

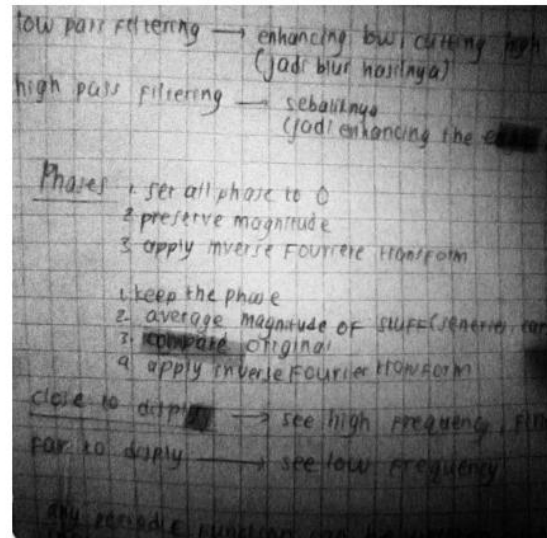
The final image look a lot less dark, making the edges of the spoon clearer, specifically the one near the handle. The edges stretches til the entire handle looks visible, unlike in the original image which kinda has it cut midway through nearing the image frame.

D. Low Contrast Image

Original Image



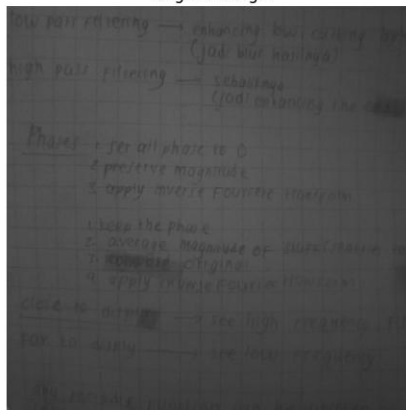
Step 2: Histogram Equalization
(PSNR: 9.66 dB)



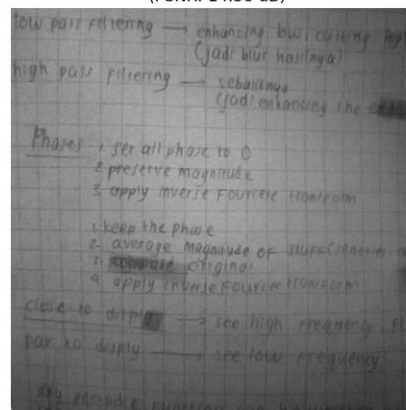
For this one, I use histogram equalization from opencv to enhance the contrast of the image by changing its distribution of its pixel intensity, and because the result looks perfect and not harsh like the previous two images, I decided to keep it as it is without any blending done.

Before this though, I did try using contrast stretching, a textbook way to improve a low contrast image. Although the resulting PSNR is higher, but as we can see from here, the result is a lot less impressive compared to the histogram equalization. The text still looks somewhat melted into the background.

Original Image



Step 1: Contrast Stretching
(PSNR: 14.38 dB)



The final resulting PSNR is 9.66 dB, which may seem low, but that is mainly because PSNR is Peak Signal to Noise Ratio that's **compared to the original image**, which makes it natural if the value would be low the more different it is to the original image. Which in this case, the original image isn't really a good comparison anyway. So I think, the low PSNR doesn't really matter if it means that the image looks visually better.

For the final image, we can see the texts more clearly, because it's a lot more contrasted with the paper background. Unlike the original that looks as if the text and the paper melts together.